C Adding 1 and 1 in modulo-2 arithmetic results in _____.
A) 1
B) 2
C) 0
D) none of the above

B A _____ error means that two or more bits in the data unit have changed.
A) double-bit
B) burst
C) single-bit
D) none of the above

D A simple parity-check code can detect _____ errors.
A) an even-number of
B) two
C) no errors
D) an odd-number of

B Checksums use _____ arithmetic.
A) two's complement arithmetic
B) one's complement arithmetic
C) either (a) or (b)
D) none of the above

C _____codes are special linear block codes with one extra property. If a codeword is rotated, the result is another codeword.
A) Non-linear
B) Convolution
C) Cyclic
D) none of the above

A In a linear block code, the _____ of any two valid codewords creates another valid codeword.
A) XORing
B) ORing

C) ANDing
D) none of the above

C In _____ coding, we divide our message into blocks, each of k bits, called ___.
A) block; blockwords
B) linear; datawords
C) block; datawords
D) none of the above

D In cyclic redundancy checking, what is the CRC?
A) The divisor
B) The quotient
C) The dividend
D) The remainder

B In _____ error correction, the receiver asks the sender to send the data again.
A) backward
B) retransmission
C) forward
D) none of the above

C 10
In _____ error correction, the receiver corrects errors without requesting retransmission.
A) backward
B) onward
C) forward
D) none of the above

C In modulo-2 arithmetic, _____ give the same results.
A) addition and multiplication
B) addition and division
C) addition and subtraction
D) none of the above

C In modulo-2 arithmetic, we use only _____.
A) 1 and 2
B) 0 and 2
C) 0 and 1

D) none of the above

A

In modulo-2 arithmetic, we use the _____ operation for both addition and subtraction.
A) XOR
B) OR
C) AND
D) none of the above

B

Speaker Independent (SI) speech recognition systems are characterized by:
A) Recognition of speech from a single user
B) Training on multiple speakers to recognize speech from any user
C) Recognition of predefined phrases only
D) Use in highly controlled environments

C

Speech recognition systems perform better when:
A) They are exposed to multiple speakers without prior training
B) They operate in noisy environments
C) They are designed to recognize a single speaker or a small group of speakers
D) The speakers use highly technical vocabulary

C

What is data compression primarily used for?
A) Enhancing data quality
B) Increasing the number of bits
C) Reducing the number of bits for transmission or storage
D) Encrypting data

C

What is the function of the Charge Coupled Device (CCD) in cameras?
A) To process audio signals
B) To store digital images
C) To convert light into electrical signals
D) To enhance digital image resolution

C

What is the major disadvantage of digital transmission compared to analog transmission?
A) It requires less bandwidth.
B) It has poorer noise immunity.
C) It requires greater bandwidth.

D) It is more susceptible to eavesdropping.

B

What is the primary function of Run-length encoding in data compression?
A) To replace sequences of zeros with non-zero values
B) To replace repeating occurrences of symbols with a single occurrence followed by the count
C) To replace repeating occurrences of symbols with a dictionary of words
D) To encrypt data before transmission

C

20
What is the primary purpose of the Huffman coding technique?
A) To encrypt data.
B) To assign shorter codes to less frequently occurring symbols.
C) To assign shorter codes to more frequently occurring symbols.
D) To convert analogue signals to digital signals.

C

What is the primary use of image enhancement in image processing?
A) To compress images to reduce file size
B) To interpret the content of an image
C) To improve the visual quality of images
D) To generate new image data

B

What is the purpose of quantization in the JPEG compression process?
A) To convert RGB to YCbCr
B) To reduce the size of the image by lowering the precision of the image data
C) To increase the brightness and contrast of the image
D) To encrypt the image data

C

What is the role of quantization in the PCM?
A) It converts the continuous signal to a discrete time signal.
B) It encodes the quantized values into bit streams.
C) It maps infinite amplitude values to a finite set of values.
D) It compresses the digital signal.

A

What is the significance of entropy in information theory?
A) It indicates the average rate of information produced by a source.
B) It represents the error rate in a communication channel.
C) It measures the bandwidth of the transmission medium.
D) It determines the power of the transmitted signal.

B

What is the typical function of source coding in digital communication systems?
A) Modulation of signals
B) Compression and formatting of data
C) Decryption of information
D) Frequency spreading

A

What type of data compression does Lempel Ziv (LZ) encoding utilize?
A) Lossless compression
B) Lossy compression
C) Encryption compression
D) Non-binary compression

C

Which compression type ensures that the original data can be perfectly reconstructed?
A) Lossy compression
B) Transformative compression
C) Lossless compression
D) Adaptive compression

C

Which compression would you choose to ensure no loss of data?
A) JPEG
B) MPEG
C) LZ encoding
D) MP3

B

Which frequency range is considered essential for commercial telephonic communication?
A) 0 Hz to 5 MHz
B) 300 Hz to 3400 Hz
C) 400 Hz to 700 Hz
D) 5 kHz to 15 kHz

A

30
Which method is a type of lossless data compression?
A) Run-length encoding
B) Quantization
C) Sampling

D) Modulation

B

Which of the following is a limitation of physical transmission channels?
A) Enhanced signal strength
B) Noise
C) Reduced cost of transmission
D) Increased data security

D

C

Which of the following is NOT a type of redundancy found in images?
A) Interpixel redundancy
B) Coding redundancy
C) Encryption redundancy
D) Interband redundancy

✓ Four primary types of redundancy that can be found in images are:

1. *Coding* 编码
2. *Interpixel* 像素带
3. *Interband* 间带
4. *Psychovisual redundancy*

C

Which of the following sources of information is characterized by a bandwidth requirement of about 15 kHz?
A) Speech
B) Data
C) Music
D) Video

C

Which type of redundancy does JPEG compression exploit?
A) Temporal
B) Spatial
C) Frequency
D) Quantitative

B

What technique is employed to achieve message confidentiality?
A) Cryptanalysis 破译密码
B) Cryptography 形成密码
C) Steganography
D) Authentication

A

In the context of cryptography, what does the term "cipher" refer to?
A) The process of converting plaintext to ciphertext
B) The algorithm used for transforming plaintext to ciphertext
C) A tool used to crack encrypted messages
D) The key used in the encryption process

B _____ means concealing the contents of a message by enciphering.
A) Steganography
B) Cryptography
C) Compressing
D) Authentication

B _____means concealing the message by covering it with something else.
A) Cryptography
B) Steganography
C) Compressing
D) Authentication

B _____means that the data must arrive at the receiver exactly as they were sent.
A) Nonrepudiation
B) Message integrity
C) Authentication
D) Secrecy

D 40
_____means that the sender and the receiver expect confidentiality.
A) Nonrepudiation
B) Integrity
C) Authentication
D) Encryption and decryption

C Three security goals are _____.
A) confidentiality, cryptography, and nonrepudiation
B) confidentiality, encryption, and decryption
C) confidentiality, integrity, and availability
D) confidentiality, denial of service, and masquerading

B What cryptographic method involves rearranging the characters of plaintext?
A) Substitution cipher
B) Transposition cipher
C) Symmetric encryption
D) Asymmetric encryption